SAT Math Mastery · Session 3

Nonlinear Equations and Systems

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics solving quadratics: factoring, completing the square, discriminant, sum and product of roots · line and parabola, line and circle systems · number of solutions and unknown constants · rational equations and extraneous solutions · exponential equations · absolute value, radical and higher-degree equations

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. Medium · No calculator needed

$$\frac{1}{x^2+10x+25}=4$$

Which of the following could be the value of $x+5$?

(A) $\frac{1}{2}$ (B) $\frac{5}{2}$

(C) $\frac{9}{2}$ (D) $\frac{11}{2}$

2. Hard · No calculator needed

In the xy-plane, the graphs of $y=x^2-6x+8$ and $y=2x-4$ intersect at two points. What is the sum of the x-coordinates of these two points?

3. Hard · No calculator needed

In the xy-plane, the graphs of $y = x^2 - 4x - 5$ and $y = 3x + 1$ intersect at two points. What is the product of the x-coordinates of these two points?

4. Hard · No calculator needed

In the xy-plane, the graphs of $y = x^2 - x - 14$ and $y = 2x - 4$ intersect at two points. What is the positive difference between the x-coordinates of these two points?

5. Hard · No calculator needed

Suppose m and n are positive integers such that $mn > 20$ and $12mn$ is a perfect square. What is the least possible value of mn?

6. Hard · No calculator needed

If $x^2 - 8x + 5 = 0$ has roots r and s, what is the value of $r^2 + s^2$?

7. HARD · NO CALCULATOR NEEDED

What is the positive solution to the equation $x^2 - 6x - 7 = 0$, found by completing the square?

8. HARD · NO CALCULATOR NEEDED

If $y = x^2 - 7$ and $x + y = 5$, what is the value of x^2y^2 for the solution (x, y) of this system with $x > 0$?

9. HARD · NO CALCULATOR NEEDED

A taxi company uses the formula $10p = q + 25r$, where p is the fare per passenger, in dollars; q is a fixed service charge, in dollars; and r is the distance traveled, in kilometers. Which equation correctly expresses p in terms of q and r?

(A) $p = \dfrac{q}{10} + 25r$

(B) $p = q + \dfrac{25r}{10}$

(C) $p = 10\,(q + 25r)$

(D) $p = \dfrac{q + 25r}{10}$

10. HARD · NO CALCULATOR NEEDED

A company's monthly profit, in thousands of dollars, is modeled by $P(x) = -2x^2 + 40x - 150$, where x is the price of the product in dollars. For what price x, other than $x = 5$, does the company break even (that is, $P(x) = 0$)?

11. HARD · NO CALCULATOR NEEDED

A system of equations is given below, where x represents the number of years since a company's founding and y represents its annual revenue, in millions of dollars.

$$y = x^2 - 5x + 20$$

$$y = 3x + 8$$

For how many values of x, with $x \geq 0$, are the two models equal?

(A) 0

(B) 1

(C) 2

(D) 3

12. HARD · NO CALCULATOR NEEDED

$$x^2\,(x + 5)\,(x - b) = 0$$

In the given equation, b is a positive constant. The sum of the solutions of the equation is 7. What is the value of b?

13. HARD · NO CALCULATOR NEEDED

If the graphs of $y = x^2 + 1$ and $y = -5x + p$ intersect at the point $(2, r)$, what is the value of $\frac{p}{r}$?

14. HARD · NO CALCULATOR NEEDED

The functions f and h are defined by $f(x) = x^2 + 3x - 40$ and $h(x) = x^2$. If $3h(x) = f(2x)$, what is the positive value of x?

15. HARD · NO CALCULATOR NEEDED

Let $f(x) = ax^3$ and $g(x) = bx^5$ for any value of x. If a and b are positive constants, for how many values of x is $f(x) = g(x)$?

(A) None

(B) One

(C) Two

(D) Three

16. HARD · NO CALCULATOR NEEDED

If $(x + 3)^2 = 1$ and $(x - 4)^2 = 36$, what is the value of x?

17. HARD · NO CALCULATOR NEEDED

What value of x satisfies the equation below?

$$\frac{x}{x-3} + \frac{2}{x} = \frac{3}{x-3}$$

18. HARD · NO CALCULATOR NEEDED

What value of x satisfies the equation $2^{3x} \cdot 4^{x+1} = 8^{x+3}$?

19. HARD · NO CALCULATOR NEEDED

$$y = x^2 - 4x + 1$$
$$y = 2x - 8$$

How many solutions does the system above have?

(A) Zero

(B) Exactly one

(C) Exactly two

(D) Infinitely many

20. HARD · NO CALCULATOR NEEDED

The equation $2x^2 - 12x + k = 0$, where k is a constant, has two real solutions. One solution is twice the other solution. What is the value of k?

21. HARD · NO CALCULATOR NEEDED

For a real number x, $4^x - 2^{x+2} - 32 = 0$. What is the value of 2^{x+1}?

(A) -8

(B) 8

(C) 16

(D) 32

22. HARD · DESMOS

The equation $|x + 6| - 7 = \dfrac{x}{3} + c$, where c is a constant, has exactly one solution x. What is the value of $2c + x$?

(A) -11

(B) -6

(C) -5

(D) -16

23. Hard · No calculator needed

In the xy-plane, the line $y = mx - 5$ intersects the parabola $y = x^2 - 6x + 11$ at exactly one point, where m is a negative constant. What is the value of m?

(A) 14

(B) -14

(C) -2

(D) 2

24. Hard · Desmos

$$x^2 + y^2 = 65$$

$$x - y = 7$$

The solution (a, b) to the system of equations above satisfies $a > 0$. What is the value of $a + 3b$?

25. Hard · Desmos

In the xy-plane, the graphs of $y = x^2 + 6x + 4$ and $y = mx - 4$ have no point in common, where m is a constant. What is the sum of all integer values of m?

26. Hard · Desmos

$$2^x = 3x + k$$

In the equation above, k is a constant. What is the smallest integer value of k for which the equation has two distinct real solutions?

(A) -2

(B) -1

(C) 0

(D) -3

27. Hard · No calculator needed

For an integer k, the equation

$$9 - 3\,|2x - 8| = 1 - \frac{5k}{2}$$

has no real solution x. What is the greatest possible value of k?

28. Hard · Desmos

For a constant k, the equation

$$2\sqrt{x+1} + 2\sqrt{15-x} = k$$

has exactly one real solution. What is the value of k?

(A) 16

(B) $2 + 2\sqrt{15}$

(C) 8

(D) $8\sqrt{2}$

29. HARD · NO CALCULATOR NEEDED

$$x^2 + y^2 = 45$$
$$y = mx + 15$$

In the system above, m is a negative constant. If the system has exactly one solution (x, y), what is the value of x?

(A) -3 (B) 3

(C) 6 (D) -6

30. HARD · NO CALCULATOR NEEDED

$$y = 3x^2 + 4$$
$$y = 1 - kx$$

In the system above, k is a constant. For which of the following values of k does the system have two distinct solutions (x, y) whose x-coordinates are both negative?

(A) 6 (B) 7

(C) -7 (D) 2

31. HARD · DESMOS

For how many integer values of k with $-10 \leq k \leq 10$ does the equation

$$\sqrt{x+k} = x - 2$$

have exactly one real solution?

32. HARD · NO CALCULATOR NEEDED

In the xy-plane, the line $y = kx+4$, where k is a constant, is tangent to the parabola $y = x^2+4x+13$ for exactly two values of k. For each of these values the line touches the parabola at one point. What is the greatest y-coordinate of these two points of tangency?

(A) 10

(B) 13

(C) 34

(D) 3

33. HARD · DESMOS

For a real number k, consider the equation

$$x - 4\sqrt{x} = k.$$

For how many integer values of k from -10 to 10, inclusive, does the equation have exactly one real solution x?

34. HARD · NO CALCULATOR NEEDED

$$x^3 + (3-k)x^2 - (10+3k)x + 10k = 0$$

In the equation above, k is a constant. The equation has exactly two distinct real solutions for exactly two values of k. What is the sum of these two values of k?

35. HARD · DESMOS

$$|2x - k| + k = x + 6$$

In the equation above, k is a constant. The equation has exactly one solution. What is the value of k?

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. Medium · No calculator needed

Emma is 4 years older than Grace. The square of Emma's age, in years, is 64 greater than the square of Grace's age. What is the sum, in years, of Emma's and Grace's ages?

37. Medium · No calculator needed

As part of a tourism marketing campaign, the number of visitors (in thousands) x weeks after the campaign begins is modeled by $v = x^2 + 2x - 3$. A separate baseline estimate predicts $v = -3$ for the same period. For how many nonnegative values of x are the two estimates equal?

(A) 0

(B) 1

(C) 2

(D) 3

38. Medium · No calculator needed

What is the positive solution to the equation $x^2 - 5x - 24 = 0$?

39. Medium · No calculator needed

What are the solutions to the equation $2x^2 - 5x - 3 = 0$?

(A) $x = -\frac{1}{2}$ and $x = 3$

(B) $x = \frac{1}{2}$ and $x = -3$

(C) $x = -\frac{3}{2}$ and $x = 1$

(D) $x = \frac{3}{2}$ and $x = -1$

40. Medium · No calculator needed

If x and y are positive integers such that $x^2 - y^2 = 45$ and $x - y = 5$, what is the value of $x + y$?

(A) 7

(B) 9

(C) 11

(D) 13

41. Medium · No calculator needed

An integer n is subtracted from its square. Which of the following could NOT be the result?

(A) 0

(B) An even integer

(C) An odd integer

(D) A positive integer

42. Medium · No calculator needed

For which value of k does the equation $x^2 + 6x + k = 0$ have exactly one real solution?

(A) 3

(B) 6

(C) 9

(D) 12

43. Medium · No calculator needed

What is the solution set of the equation $\dfrac{x}{x-2} = \dfrac{2}{x-2} + 3$?

(A) $x = 2$ only

(B) $x = 4$ only

(C) $x = 2$ and $x = 4$

(D) There is no solution.

44. Medium · No calculator needed

Which inequality is equivalent to $8x - 12y > 20$?

(A) $2x - 3y > 5$

(B) $3x - 2y > 5$

(C) $4x - 3y > 5$

(D) $2y - 3x > 5$

45. Medium · No calculator needed

How many distinct real solutions does the equation $4x^2 - 20x + 25 = 0$ have?

(A) Exactly one

(B) Exactly two

(C) Infinitely many

(D) Zero

46. Medium · No calculator needed

In the system of equations below, x and y satisfy both equations.

$$y = 5x + 2$$

$$y = 5x^2 + 2$$

Which ordered pair (x, y) is a solution to this system?

(A) $(0, 0)$ (B) $(0, 5)$

(C) $(1, 5)$ (D) $(1, 7)$

47. Medium · No calculator needed

$$7|x| = 63$$

What is the positive solution to the given equation?

48. Medium · No calculator needed

What is the sum of the y-coordinates of the solutions to the system below?

$$y = x^2 - 4x$$

$$y = 2x - 5$$

(A) -2 (B) 0

(C) 2 (D) 4

49. Medium · No calculator needed

The height of a triangle is 4 cm more than its base. If the area of the triangle is 30 cm^2, what is the length of the base, in cm?

(A) 4

(B) 5

(C) 6

(D) 8

50. Medium · No calculator needed

What is the solution to the equation $\dfrac{x+5}{x-1} = 3$, for $x \neq 1$?

51. Medium · No calculator needed

The curve $y = \dfrac{x^2}{3}$ and the line $y = \dfrac{2x}{3}$ intersect at the origin and at the point (a, b). What is the value of b?

52. Medium · No calculator needed

For what values of x is the expression $2x^2 - 2x - 24$ equal to 0?

(A) $x = -4,\ x = 3$

(B) $x = 4,\ x = -3$

(C) $x = 4,\ x = 3$

(D) $x = -4,\ x = -3$

53. Medium · No calculator needed

If $(x-3)(x+7)=-25$, what is the value of x?

54. Medium · No calculator needed

The square of a positive number is 12 more than 4 times the number. What is the number?

55. Medium · No calculator needed

$$xy = 48$$

$$3x - y = 0$$

If $(x,\ y)$ is a solution to the system above and $x > 0$, what is the value of y?

56. Medium · No calculator needed

If $2^{x+3} - 2^x = 56$, what is the value of x?

57. Medium · No calculator needed

What is the product of the positive solutions to the equation $x^4 - 13x^2 + 36 = 0$?